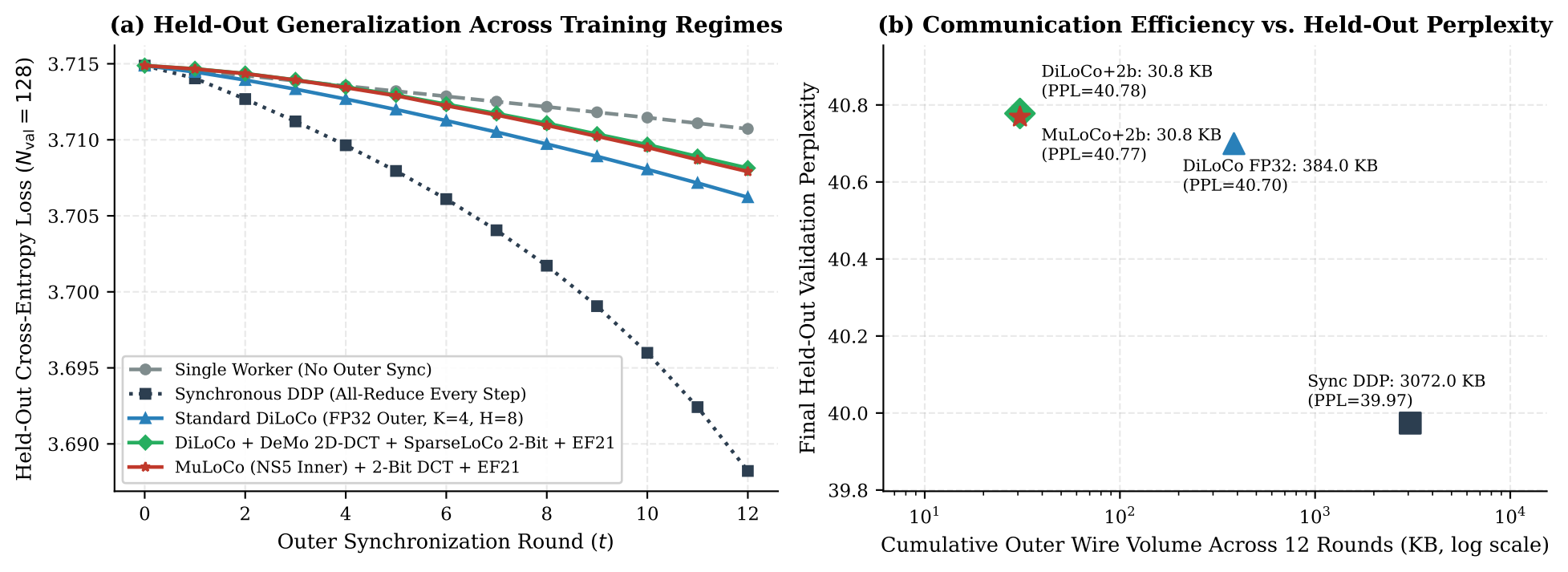
(a) Held-Out Generalization Across Training Regimes
Held-Out Cross-Entropy Loss ($N_{val} = 128$)
3.715
3.710
3.705
3.700
3.695
3.690
0
2
4
6
8
10
12
Outer Synchronization Round (t)
Single Worker (No Outer Sync)
Synchronous DDP (All-Reduce Every Step)
Standard DiLoCo (FP32 Outer, K=4, H=8)
DiLoCo + DeMo 2D-DCT + SparseLoCo 2-Bit + EF21
MuLoCo (NS5 Inner) + 2-Bit DCT + EF21
(b) Communication Efficiency vs. Held-Out Perplexity
Final Held-Out Validation Perplexity
40.8
40.6
40.4
40.2
40.0
39.8
10^1
10^2
10^3
10^4
Cumulative Outer Wire Volume Across 12 Rounds (KB, log scale)
DiLoCo+2b: 30.8 KB
(PPL=40.78)
MuLoCo+2b: 30.8 KB
(PPL=40.77)
DiLoCo FP32: 384.0 KB
(PPL=40.70)
Sync DDP: 3072.0 KB
(PPL=39.97)